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Condition monitoring based on modified CUSUM and EWMA control charts

CUSUM and
EWMA control
charts

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Abstract

Purpose – The application of condition-based maintenance on selected equipment can allow online monitoring using fixed, half-fixed or portable sensors. The collected data not always allow a straightforward interpretation and many false alarms can happen. The paper aims to discuss these issues.

Design/methodology/approach – Statistical techniques can be used to perform early failure detection. With the application of Cumulative Sum (CUSUM) Modified Charts and the Exponentially Weighted Moving Average (EWMA) Charts, special causes of variation can be detected online and during the equipment functioning. Before applying these methods, it is important to check data for independence. When the independence condition is not verified, data should be modeled with an ARIMA (p, d, q) model. Parameters estimation is obtained using the Shewhart Traditional Charts.

Findings – With data monitoring and statistical methods, it is possible to detect any system or equipment failure trend, so that we can act at the right time to avoid catastrophic failures.

Originality/value – In this work, an electro pump condition is monitored. Through this process, an anomaly and four stages of aggravation are forced, and the CUSUM and EWMA modified control charts are applied to test an online equipment monitoring. When the detection occurs, the methodology will have rules to define the degree of intervention.

Keywords Condition-based maintenance, ARIMA model, CUSUM chart, EWMA chart, Shewhart control chart

Paper type Research paper

1. Introduction

Corrective Maintenance is the action that restores failed equipment to its operating state (NP EN 13306:2007 Standard). Preventive maintenance – specifically on condition-based maintenance (CBM) – uses historic failure data in conjunction with optimization and statistical models to predict equipment optimal replacement period (Dias *et al.*, 2007). To identify critical elements for the explanation of failure processes, several models can be used such as the parametric regression models (Accelerated Failure Time Models) and the non-parametric models (Proportional Hazards Models) (Lawless, 1982).

Repairable naval system monitoring can be accomplished installing sensors in critical system points to collect, in real time, the values of a large number of variables characterizing equipment condition; diagnosis based on this data is a key factor for the correct application

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of the CBM concept. Given the large number of observed variables, unexpected results sometimes happen.

The use of control charts for CBM on repairable systems and equipment can allow the condition monitoring of those machines (Dias *et al.*, 2009). In this case, enough data should exist. This methodology will point to a preventive maintenance need on the affected component and in time.

In this paper, we present two types of charts that can be used for equipment monitoring: Cumulative Sum (CUSUM) (Gan, 1991) and Exponentially Weighted Moving Average (EWMA) control charts (Pereira and Requeijo, 2012). These are modified control charts that are designed and applied for the purpose of controlling water electro pump vibration control and design for this purpose.

2. Control charts

Statistical control charts are usually used in achieving control process stability, and integrating the statistical process control (SPC) systems, in products quality control.

On SPC various control charts are applied; in this paper, we propose the use of Shewhart charts in Phase 1 and the special control charts in Phase 2.

In Phase 1, two types of charts are commonly used: one to control the mean and another to control the process dispersion. In this phase, the retrospective control is done, where m samples are collected and used to estimate the process parameters; when the variation causes are common, a stable process is present. The defined parameters mean and variance will be used in Phase 2 for process monitoring. In this study, the individual observation (X) and moving range (MR) control charts are used in Phase 1.

In SPC, Phase 2 process monitoring is performed using the statistical control process. In this phase, the special control charts CUSUM and EWMA with a modification are used.

2.1 Phase 1

2.1.1 Independent data. Having enough data to estimate the process parameters, Phase 1 of SPC with Shewhart charts is implemented.

For continuous variables, the X and MR control charts are implemented. The control charts analysis must show a controlled statistical process. As for values interpretation, the inexistence of random patterns is considered (ISO 8258:1991 Standard). This is applied only if data are independent, considering $X_i = \mu + \varepsilon_i$, with ε white noise ($\varepsilon \sim N(o; \sigma_\varepsilon^2)$).

If data are independent, Phase 1 should proceed, and the Shewhart control chart is built. The upper control limit (UCL), the lower control limit (LCL) and the center line (CL) of those charts, are calculated using the formulas present in Table I.

The X and $\overline{MR}$ limits in the Table I are calculated based on the m (or $m-1$) sample statistics, as $\overline{X} = \sum_{i=1}^m X_i / m$ and $\overline{MR} = \sum_{i=1}^{m-1} MR_i / (m-1)$, where $MR_i = |X_i - X_{i-1}|$.

If the process is stabilized – under statistical control – its parameters should be estimated considering $\hat{\mu} = \overline{X}$ and $\hat{\sigma} = \overline{MR} / d_2$. The constants D_3 , D_4 and d_2 depend exclusively on the sample dimension and their values are defined on control charts factors tables for variables traditional control charts defined by Shewhart (Requeijo *et al.*, 2012).

2.1.2 Autocorrelationated data. Due to the reliability of results, Shewhart control charts are applied only with independent data. If this is not the case, two approaches are possible.

Table I.
Shewhart control
charts limits, in
Phase 1 of SPC

Chart	LCL	CL	UCL
X	$\overline{X} - 3\sigma_X$	$\overline{X}$	$\overline{X} + 3\sigma_X$
MR (moving range)	$D_3 \overline{MR}$	$\overline{MR}$	$D_4 \overline{MR}$

The first one implies the fitting of a statistical model, and with the second one, this is not necessary.

In condition control, only the first case makes sense, and the modeling for autocorrelated data is done with ARIMA models.

Like in SPC, for independent data, the techniques for the first phase are the traditional Shewhart charts, but for the residues, we apply adjustment. Based on the ARIMA model, adjusted by $AR(p)$, $MA(q)$ or $ARMA(p, q)$, the localization (μ) and dispersion (σ) parameters estimation proceeds. When there is a special cause of variation, those parameters should be replaced with the expected values for the respective instant, adjusted by the ARIMA model; the new residues should be calculated and the same for the reviewed charts (Lampreia *et al.*, 2012).

To check for a significant autocorrelation in the process, autocorrelated function (ACF) and partial autocorrelated function (PACF) are used together. In this circumstance, the most common approach is to model the process using the Autoregressive Integrated Moving Average (ARIMA) model.

To model the process using this methodology, the ARIMA (p, d, q) model that best fits data is identified comparing the estimated autocorrelation Function (EACF) with theoretical autocorrelation function (ACF), and the estimated partial autocorrelation function (EPACF) with the theoretical partial autocorrelation function (PACF).

A process follows an ARIMA (p, d, q) model if $\nabla^d X_t$ follows an ARMA (p, q) model. To see how to define an ARIMA (p, d, q) model, Pereira and Requeijo's (2012) work must be consulted.

When a satisfactory model is obtained, its residues are estimated by $e_t = X_t - \hat{X}_t$ ($\hat{X}_t$ is the expected value for the period t). Control charts are built from these residues. The mean and standard deviation are estimated from the modeled data. For the model AR (p), process parameters are estimated from the following expressions (Pereira and Requeijo, 2012):

$$\hat{E}(X) = \hat{\mu} = \bar{\xi} / \left(1 - \sum_{j=1}^p \varphi_j \right) \text{ and } \hat{VAR}(X) = \hat{\gamma}_0 = \sigma_e^2 / \left(1 - \sum_{j=1}^p \rho_j \varphi_j \right) \quad (1)$$

2.2 Phase 2

In Phase 2, the modified CUSUM or EWMA control charts are implemented. These charts are considered special charts because of their capability to detect small changes and special causes (Perry and Pignatiello, 2011).

CUSUM charts are based on "memory" statistics as opposed to Shewhart charts. The analysis of an out of control situation that involves the values at the present time and the value at the previous time (Pereira and Requeijo, 2012).

The EWMA control charts are sensitive to small changes in the process and they are used to forecast the observation in the next period, it was introduced by Roberts (1959).

The main objective in this work is to predict, online, the instant where a repairable system reaches a parameter value corresponding to a failure. A modification in CUSUM and EWMA charts designed to detect high vibrations as soon as possible is presented. Two alternatives are considered; one for independent and another for autocorrelated data.

2.2.1 Independent data. Modified CUSUM. "Modified CUSUM" chart is built on cumulative sum – C (Sibanda and Sibanda, 2007) defined by:

$$C_t = \max(0, C_{t-1} + (Z_t - k)); \quad C_0 = 0 \quad (2)$$

where $Z_t = ((\bar{X}_t - T_L) / \sigma_{\bar{X}})$, $\sigma_{\bar{X}} = \sigma / \sqrt{n}$, $\Delta = \delta \sigma_{\bar{X}}$, $k = \delta/2$ e $T_L = (T_L)_{\text{Standard}} - \Delta_s$ and $\Delta_s = \delta_1 \sigma$, where δ_1 is a constant.

In this definition, $\bar{X}_t$ is the sample mean at t , T_L is the maximum admissible vibration, σ the process standard deviation, n the sample dimension, Z_t is the reduced form of $\bar{X}_t$, k the reference value and Δ_s the safety factor (Barbosa, 2012).

The "Modified CUSUM" has two limits; one works like an Alert Level (AL), and the second is the real Upper Control Level (UCL). The AL and UCL calculations are based on Gan (1991) abacus, in function of Average Run Length (ARL) value for both situations and the reference value k . In this study, a significance level $\alpha = 1$ percent (ARL = 100) will be considered to define AL and $\alpha = 0.2$ percent (ARL = 500) to define UCL.

In Point 3, a methodology to apply the Modified CUSUM charts to independent and autocorrelated data are presented.

Modified EWMA. The Modified EWMA is defined by the exponentially weighted variable, E , defined by the following equation (Zou and Tsung, 2010):

$$E_t = \max(0, (1-\lambda)E_{t-1} + \lambda(\bar{X}_t - T_L)), \text{ com } 0 < \lambda < 1 \quad (3)$$

where, $\sigma_{\bar{X}} = \sigma_e / \sqrt{n}$, $\Delta = \delta \sigma_{\bar{X}}$, $T_L = (T_L)_{\text{Standard}} - \Delta_s$ and $\Delta_s = \delta_1 \sigma$, with δ_1 as a constant (Sereland Moskowitz, 2008).

In these equations, $\bar{X}_t$ is the sample mean at t , T_L the maximum vibration admissible, σ_E the variable variance from E , n the sample dimension, λ the weight constant ($0 \leq \lambda \leq 1$) and Δ_s the safety factor (Lampreia *et al.*, 2012). Alert level and UCL for these charts are given by the following equations (Crowder and Hamilton, 1992):

$$LA_E = K_1 \sigma_E \quad (4)$$

$$LSC_E = K_2 \sigma_E \quad (5)$$

where:

$$\sigma_E = \sigma_{\bar{X}} \sqrt{\frac{\lambda}{2-\lambda}} \quad (6)$$

K_1 and K_2 are the values determined by the values of λ and ARL (for AL and UCL).

The Modified EWMA has two control limits like CUSUM charts, the first one works like a warning (alert) (AL), and the second is the UCL. To determine the AL and UCL values the Crowder (1987) abacus is used, in function of the ARL value and the mean change, δ . And then to determine K the ARL and the weighted constant, λ , are used. The values of α and ARL are the same as in the CUSUM charts.

2.2.2 Autocorrelated charts. When data are autocorrelated, the Modified EWMA chart is built on the exponentially weighted variable, E , defined by Equation (3), and the Modified CUSUM is built on the cumulative sum C defined by Equation (2), both calculated by using the prediction errors, e_t . Those prediction errors are estimated at the timer by $e_t(T) = X_{T+\tau}(T) - \hat{X}_{T+\tau}(T)$, where $X_{T+\tau}(T)$ is the value for that time, and $\hat{X}_{T+\tau}(T)$ the predicted value for time τ ; the predicted value for the present time is the same as the final value of Phase 1.

In the calculation of the value of E , we use the prevision values, so it is necessary to correct the calculation of T_L value, using $T_L = (T_L)_{\text{Norma}} - \mu - \Delta_s$, where μ is the process mean, estimated in Phase 1 (Requeijo *et al.*, 2012; Lampreia *et al.*, 2012).

3. Monitoring by CUSUM and EWMA methodology

The proposed methodology is to be applied to vibration monitoring of repairable systems using statistical control charts, considering both independent and autocorrelated data. Specifically, the following steps must be taken:

CUSUM and
EWMA control
charts

- (1) Collect data, defining the sampling rate.
- (2) With the equipment in a normal state (without anomalies), check for data independence, using EACF and EPACF.
- (3) For independent data, build the traditional control charts (X and MR charts), verify the process stability, the Normality condition, and estimate its parameters –vibration mean and standard deviation.
- (4) For autocorrelated data, fit an ARIMA Model.
- (5) Build e -MR Charts from the residues, verify the process stability and Normality condition of residues; estimate the system vibration process parameters (mean and standard deviation).
- (6) To monitor vibration process, build the Modified CUSUM and EWMA charts for independent and autocorrelated data, using the values collected after the anomaly introduction. When data are not independent (autocorrelated data) prediction errors are applied, instead of the original data:
 - Define, based on the ISO 2372 standard, the vibration limit value to take a maintenance action.
 - Define the change in the vibration system mean that must be detected.
 - Estimate the two control limits – AL and UCL – for the vibration mean value.
 - Fix the intervention rules:
 - Proceed to an inspection, when eight consecutive observations are above the AL.
 - Proceed to a maintenance intervention when five observations are above the UCL.

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4. Case study

This case study is based on condition monitoring of an electro pump shown in Figure 1. Point 1 is the place where vibration data were collected (Figure 2).

Collected data were global vibration values (RMS). The measurements were made at five minutes interval, after a stabilization running period of 30 minutes. Data are expressed in velocity units (mm/s).

4.1 Phase 1

In total, 241 samples of individual observations were collected ($n = 1$) with the electro pump in a good working condition (normal state).

The first step was parameters estimation, Phase 1 of SPC. In first place we verified the independence condition using the FACE and FACPE analysis. For those we used Statistica software. This analysis showed the presence of a significant autocorrelation, so the Shewhart charts could not be applied directly.

To identify the best ARIMA (p, d, q) model and adjust to the data, we compared the EACF and EPACF results with the ACF and PACF, respectively. In this case, that comparison suggested an AR (2) model ($\hat{\xi} = 0.2369$; $\hat{\phi}_1 = 0.3055$; $\hat{\phi}_2 = 0.2409$).

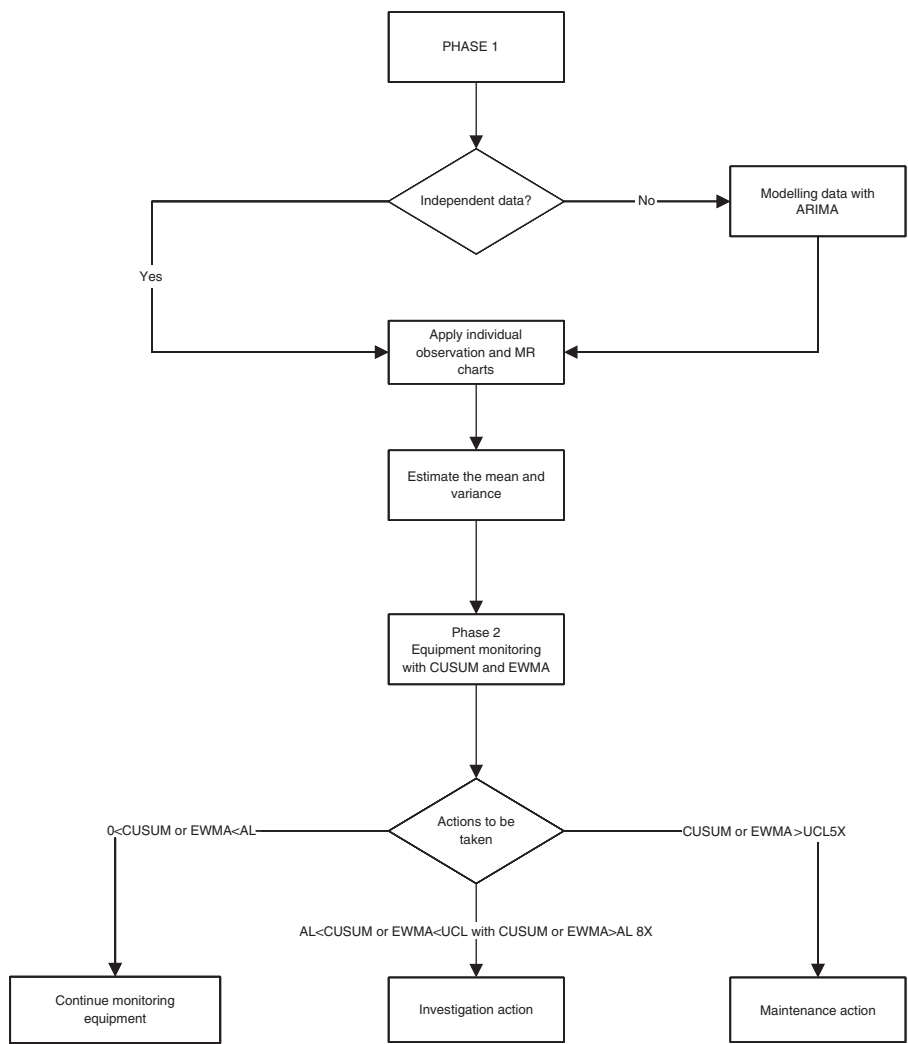


Figure 1.
Methodology
flowchart

Using Statistica again, the residues were calculated and their independence was checked, and *e*-MR charts (residues and moving average) were built. The obtained charts showed the existence of a special cause in the 45th observation, which is above the control limit of the MR chart. Because we were using the residues, this point was not eliminated but replaced with the expected value.

After this replacement, it was necessary to readjust the model to the predicted value, so we obtained the EACF and EPACF shown in Figures 3 and 4, respectively. Those graphics revealed the existence of a significant autocorrelation.

Then, the new ARIMA model was adjusted, and new parameters were estimated (Tables II and III) and a control chart with residues was built. The revised *e*-MR is presented in Figure 5.

From the analysis of Figure 4, no special cause of variation was detected.

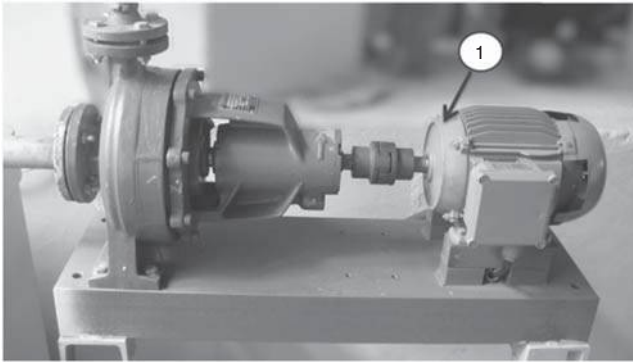


Figure 2.
Electro pump

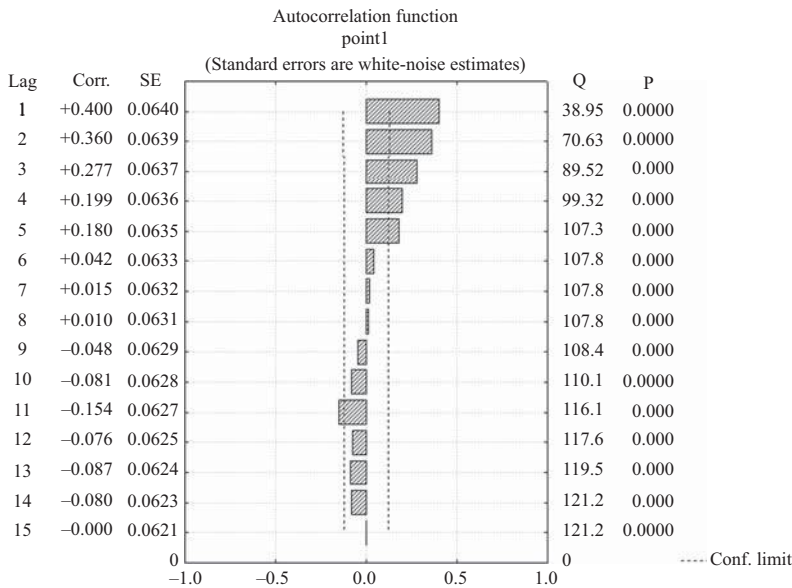


Figure 3.
EACF for data
collected on Point 1

The normality of data were checked using Kolmogorov-Smirnov test ($D_{\hat{A}_{Cr\acute{U}tico}} = (0.886/\sqrt{N}) = (0.886/\sqrt{250}) = 0.0571$ for $\alpha=5$ percent); for Point 1: $d=0.04494$. Since $d < D_{\hat{A}_{Cr\acute{U}tico}}$ the normality condition was not rejected (see Figure 6).

Having verified the data stabilization, we proceeded to parameters estimation phase, as shown in Table III.

For electro pump condition monitoring, the modified CUSUM and EWMA control charts were applied to the predicted errors, because those charts allow the detection of small changes in the vibration mean value, and also in the equipment running pattern.

According to ISO 2372: 2003 standard which gives 1.12 mm/s (RMS) as the allowed vibration limit for an equipment of class I (like the electro pump under study with an electrical engine of 1.5 KW), the $(T_L)_N$ value was defined. Consequently, we had $T_L = 1.12 - 0.517 - 0.04 = 0.56$ mm/s.

Figure 4.
EPACF for data
collected on Point 1

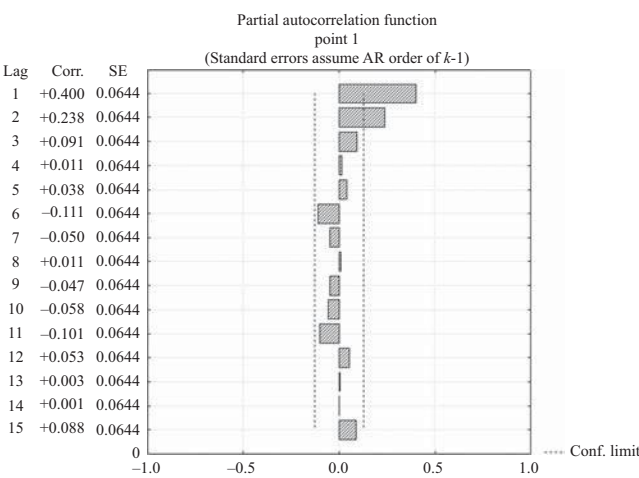


Table II.
Parameters revision
for the model adjusted
to Point 1

$\hat{\xi}$	$\hat{\phi}_1$	$\hat{\phi}_2$
0.2369	0.3055	0.2409

Table III.
CUSUM and EWMA
control charts
estimated parameters

$\overline{MR}$	$\hat{\sigma}_e$	$\hat{\mu}$	$\hat{\sigma}$
0.04393	0.03895	0.522	0.043

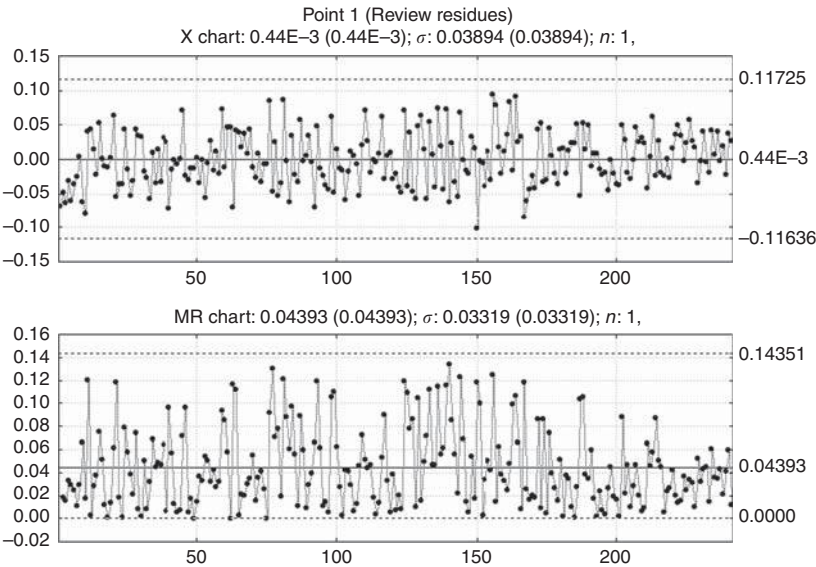


Figure 5.
 e -MR residues control
chart review for
Point 1

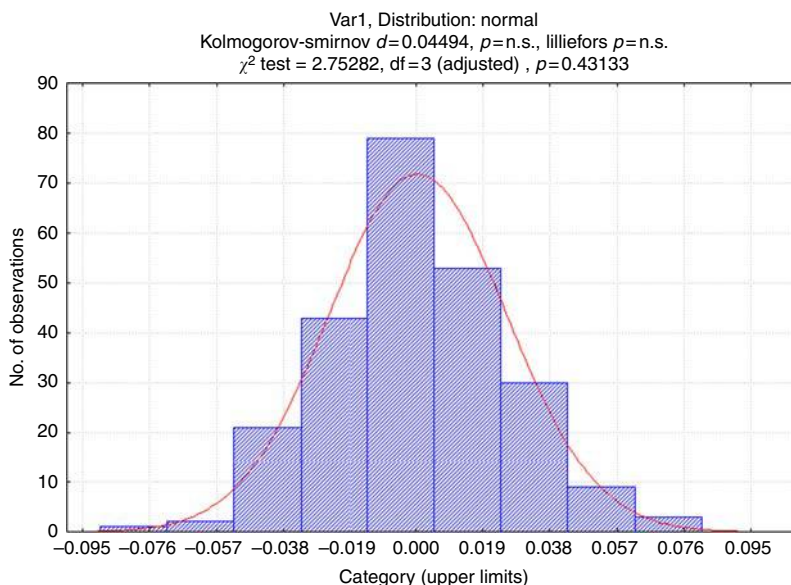


Figure 6.
Point 1 – normal
distribution

To accelerate the pump failure process, one anomaly was introduced, and four aggravation states were, successively, applied. The anomaly was the unscrewing of the support screws. After that, we collected 50 observations for each state ($n = 1$).

For this study, the need to detect ($\Delta = 0, 5\sigma_e$) was defined, however, to test the chart sensibility for different values of Δ , to the higher aggravation, the results were also represented for $\Delta = 1\sigma_e$ and for $\Delta = 1.5\sigma_e$.

4.2 Phase 2

In the second phase of this paper, an anomaly with four stages of deterioration, were introduced. The intention was to apply both charts demonstrating its effectiveness. The electro pump vibration record was obtained using portable equipment. The charts are built using Microsoft Excel.

4.2.1 Modified CUSUM chart. For the limits control charts, $\alpha = 0.1$ percent ($ARL = 100$) was used for the specification of the AL and $\alpha = 0.2$ percent ($ARL = 500$) for the UCL specification. Given the ARL values and the k reference value, it was possible using Gan (1991) abacus to specify the charts limits (h and h_1), as shown in Table IV.

Using Microsoft Excel, the Modified CUSUM control charts for four aggravations having ($\Delta = 0, 5\sigma_e$) are represented in the next figures. Because of the relevance, only the third and fourth aggravation will be represented.

Analyzing the obtained results for each aggravation, we can see that in the first one, almost no changes are detected in the mean vibration value. After the second aggravation, some changes start to occur in the chart, although none passes the alert limit (Figure 7).

		AL(h); ARL = 100	UCL(h_1); ARL = 500
k	0.25	5.51	8.5
	0.5	3.5	5.1
	0.75	2.5	3.5

Table IV.
Modified CUSUM
charts limits

Figure 7.
Modified CUSUM
chart for the 3rd
aggravation with
($\Delta = 0.5\sigma_e$)

After the third aggravation, some observations above the AL and UCL are shown. From observation 32th to 50th, all observations are above the AL, and some are even above the UCL (observation 41th to 44th and from observation 48th to 50th). According to the established rules, an action must be taken: an investigation action must take place since there are eight consecutive observations above the AL (although some observations are above the UCL, the rule to an intervention action is not verified) (Figure 8).

As soon as the 4th aggravation is introduced, we can detect significant changes in the mean vibration level. After Observation No. 25, the chart shows values close to the AL and, after 34th observations, this limit is exceeded. From the sample 39th to sample 50th, all observations are above the UCL. In the neighborhood of observation 50th an increasing trend is seen, meaning that it is expected that if no maintenance intervention action is taken, the electro pump degradation will increase and, consequently, the vibration level also.

In next two charts, referred to the 4th aggravation, we can see the difference resulting from the use of a lower Δ , $\Delta = 1\sigma_e$ and a higher one, $\Delta = 1.5\sigma_e$.

Comparing the results in Figure 9 with those shown in Figures 10 and 11, it is possible to study how Δ influences the chart sensibility. For $\Delta = 1\sigma_e$, Figure 9 shows that

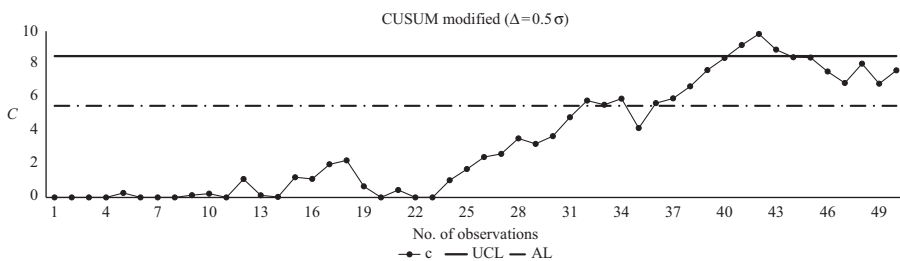


Figure 8.
Modified CUSUM
chart for the 4th
aggravation with
($\Delta = 0.5\sigma_e$)

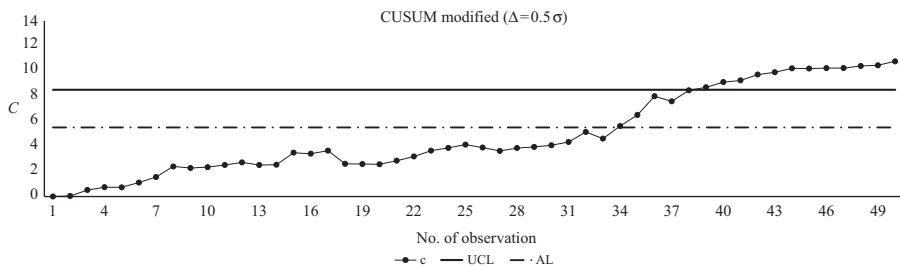
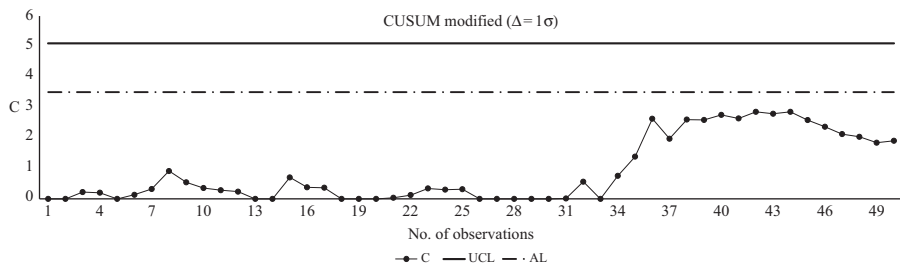


Figure 9.
Modified CUSUM
chart for the 4th
aggravation with
($\Delta = 1\sigma_e$)



for some observations, the mean vibration value is very close to AL, never exceeding it. For $\Delta = 1, 5\sigma_e$, Figure 10 shows that for some points below the AL limit, no significant changes occur.

4.2.2 Modified EWMA charts. EWMA chart was applied assuming the same conditions as for CUSUM charts.

For limit control calculations, $\alpha = 1$ percent ($ARL = 100$) is used for AL setting and $\alpha = 0.2$ percent ($ARL = 500$) for UCL definition. Given ARL value and the reference value k , it is possible, using the Crowder (1987) abacus, to specify the charts limits (K and K_1), as shown in Table V.

Examining EWMA chart, with $\Delta = 0.5\sigma_e$, for the first aggravation, we can see that no changes are detected in the vibration value. For the second aggravation, like in the CUSUM charts, some changes start to appear. After the second unscrewing, some significant changes appear in the chart but all below the AL (Figure 11).

For the third aggravation, from observation 37th to 46th, all are above the AL, and some of them – 40-43 – above the UCL. According to the established rules, the need for an anomaly action investigation occurs (Figure 12).

When the fourth aggravation is introduced, after observation 34th, the values are above the AL, and after 38th observation, all of them are above the UCL. This means that, after observation 41th, an investigation action is needed, and, after observation 43th, an intervention action is needed (Requeijo *et al.*, 2012).

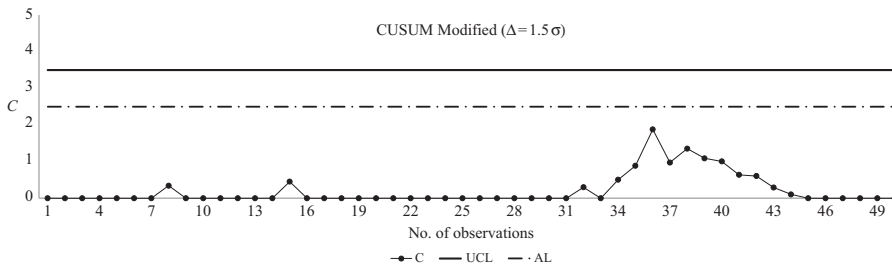


Figure 10.
Modified CUSUM
chart for the 4th
aggravation with
($\Delta = 1.5\sigma_e$)

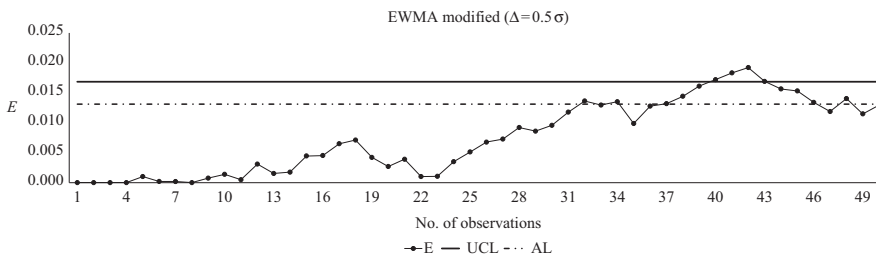


Figure 11.
Modified EWMA
chart for the 3rd
aggravation with
($\Delta = 0.5\sigma_e$)

	LA(K); ARL = 100		LSC(K1); ARL = 500
δ	1.5	2.5	3
	1	2.3	2.9
	0.5	2.1	2.7

Table V.
Modified EWMA
charts limits

Like in the fourth aggravation on the CUSUM control charts, we tested the charts for $\Delta = 1\sigma_e$ and $\Delta = 1.5\sigma_e$, and the result was the same, the charts are less sensible to that Δ values. So, to these values, the charts will not be represented. These results mean that the best analysis corresponds to $\Delta = 0.5\sigma_e$.

5. Conclusion

Given the equipment characteristics used for this work, a vibration measurement was used to monitor the equipment performance. For other kind of equipment – a diesel engine for example – different parameters can be studied (Lampreia *et al.*, 2012).

The application of control charts for online condition monitoring allows equipment monitoring, avoiding false alarms, because it shows the progress of state degradation. So, a planned maintenance according with the state, as opposed to unplanned maintenance needs, is possible.

If the data are autocorrelated, the ARIMA model should be applied.

In Phase 1, the individual observation and MR control charts should be applied to define the equipment functioning parameters; if data are independent it is applied directly, and if not, the residues should be applied on Phase 1 and the predicted values on Phase 2.

The charts sensibility depends on the Δ value. For equipments in continuous operation, a sensibility analysis is necessary to obtain the best value for Δ .

Traditional charts can also be used in Phase 2 but they are less sensible for small shifts detection.

Both Modified CUSUM and EWMA charts on Phase 2, applied to independent or autocorrelated data, are effective methods for vibration trend detection, allow in early interventions planning, before catastrophic failures.

Using SPC for equipment monitoring, considering the applied modifications, it is possible to detect incipient faults, so technicians can act before the damage occurs.

Preceding or simultaneously with this kind of investigation, other types of data analysis-thermograph or oil analysis for example – should be used to complement the study.

We believe that the control charts application is a useful method that allows knowing online equipment real state and its tendency. This probably will enhance the performance of maintenance, minimize the unplanned loss of systems operationally and eventually reduce maintenance and operability costs.

6. Future work

Test other variables simultaneously and apply the MCUSUM and MEWMA charts.

Test the flexibility concept in the charts online monitoring, for example, if there is a lack of equipment data, apply the Quesenberry control charts, and when there is enough data, change to individual observation charts for univariate study or to a T^2 charts for a multivariate study considering Phase 1.

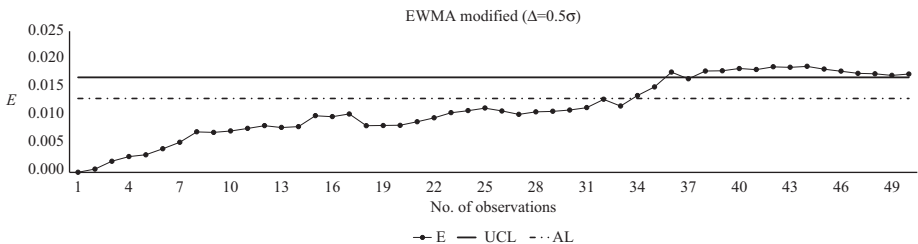


Figure 12.
Modified EWMA
chart for the 4th
aggravation with
($\Delta = 0.5\sigma_e$)

Test the charts applicability considering the use of other equipment parameters except the vibration (e.g. temperature, pressure, electrical and ultrasonic measures).

Develop software to allow control charts appliance flexibility in online monitoring.

CUSUM and
EWMA control
charts

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